

CHEETI HARI CHANDRA PRASAD RAO

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WORK EXPERIENCE

Metry AI Inc. — Backend Engineer → Founding Engineer

03/2026 – Present

Python, TypeScript, Node.js, PostgreSQL, REST APIs, Microservices, Docker, GitHub Actions, Grafana

- Own backend architecture and REST API development for an AI-powered operations platform, carrying microservices and data models from design through production support.
- Cut API response times **33%** by reworking database indexing, caching, and query paths to eliminate redundant PostgreSQL reads.
- Integrated LLM APIs and rebuilt prompt workflows behind structured testing and evaluation, improving response quality and consistency **25%** across platform features.
- Instrumented API performance, service health, and production failures in Grafana, shortening the path from a failure firing to a diagnosed root cause.
- Promoted from Backend Engineer to the founding engineering team, working largely async with a Canada-based team across a **~10-hour** offset on architecture, CI/CD, and reliability calls.

PROJECTS

macOS Troubleshooting Agent — Agentic RAG System

CODE github.com/Harichandra1/macos-agent | LIVE macos-agent-hari.duckdns.org

Python, LangGraph, FastAPI, Qdrant, Llama 3.3 70B, PostgreSQL, Docker, Prometheus, Grafana, Oracle Cloud

- Engineered an agentic RAG system for macOS troubleshooting on **LangGraph, FastAPI, Qdrant, and Llama 3.3 70B**, with a branching workflow for retrieval, clarification, diagnostics, and verification.
- Grounded the agent in a domain knowledge base built from **9 custom ingestion pipelines**, with web fallback gated on retrieval confidence and difficulty-aware solution triage.
- Built an **LLM-as-judge** evaluation harness scoring correctness, actionability, grounding, and troubleshooting depth, wired to a regression gate that blocks deploys on degradation.
- Benchmarked against GPT-4o and raw Llama-70B on multi-turn cases: **50% resolution vs. 25%** for GPT-4o (**n=4** — directional, not a general performance claim).
- Productionized with Google OAuth/JWT, PII and prompt-abuse guardrails, LangGraph memory persisted in PostgreSQL, and **Groq→NVIDIA→OpenRouter** provider failover.
- Deployed via Docker Compose on Oracle Cloud with Caddy TLS, GitHub Actions CI, Alembic migrations, and Prometheus/Grafana/Loki.

Deep Research Agent

Python, OpenAI Agents SDK, Multi-Agent Systems, AsyncIO, Pydantic, Gradio

- Built a multi-agent research system that turns a broad query into a planned investigation — web search, evidence synthesis, and a written report delivered by email.
- Orchestrated **Planner → parallel Search → Writer → Email** agents with AsyncIO fan-out, completing end-to-end runs in **~20–60 seconds**.
- Enforced structured handoffs between agents with Pydantic schemas and tool calling on the **OpenAI Agents SDK**, replacing single-pass responses with a staged pipeline.

TECHNICAL SKILLS

LLM & Agents: LangGraph, OpenAI Agents SDK, LangChain, multi-agent orchestration, tool calling, agentic RAG, LLM-as-judge evaluation, guardrails

Backend: Python, FastAPI, TypeScript, Node.js, Express.js, REST APIs, microservices, AsyncIO, Pydantic, SSE

Data & Retrieval: PostgreSQL, Qdrant, FAISS, Pinecone, SQLAlchemy, Alembic, embeddings, vector search

Infra & Observability: Docker, Docker Compose, GitHub Actions CI/CD, Oracle Cloud, Caddy, Git, Prometheus, Grafana, Loki

EDUCATION

Manipal Institute of Technology — B.Tech, Computer Science, 2021–2025 · Karnataka, India

CERTIFICATIONS

AWS Certified Cloud Practitioner — credly.com/badges/1980ca54